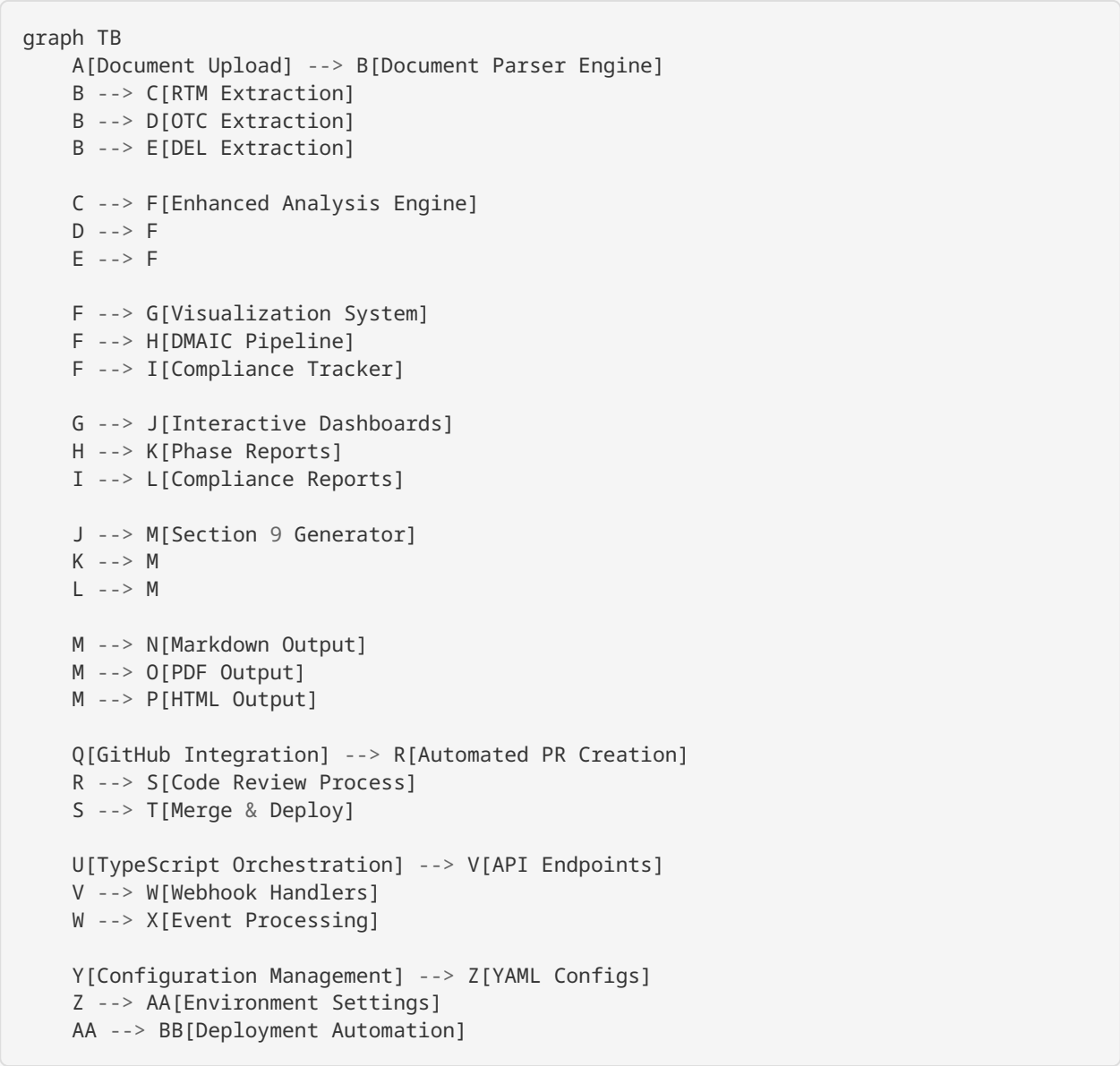


# Workflow Visualization with Mermaid Diagrams

## System Architecture Overview



## DMAIC Process Flow

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```

flowchart TD
    Start([Project Initiation]) --> Define[Define Phase]

    Define --> D1[Problem Statement]
    Define --> D2[Project Charter]
    Define --> D3[SIPOC Diagram]
    Define --> D4[Stakeholder Analysis]

    D1 --> DefineGate{Define Gate Review}
    D2 --> DefineGate
    D3 --> DefineGate
    D4 --> DefineGate

    DefineGate -->|Pass| Measure[Measure Phase]
    DefineGate -->|Fail| DefineRework[Rework Define]
    DefineRework --> Define

    Measure --> M1[Data Collection Plan]
    Measure --> M2[Baseline Measurements]
    Measure --> M3[MSA - Measurement System Analysis]
    Measure --> M4[Process Mapping]

    M1 --> MeasureGate{Measure Gate Review}
    M2 --> MeasureGate
    M3 --> MeasureGate
    M4 --> MeasureGate

    MeasureGate -->|Pass| Analyze[Analyze Phase]
    MeasureGate -->|Fail| MeasureRework[Rework Measure]
    MeasureRework --> Measure

    Analyze --> A1[Root Cause Analysis]
    Analyze --> A2[Statistical Analysis]
    Analyze --> A3[Hypothesis Testing]
    Analyze --> A4[Gap Analysis]

    A1 --> AnalyzeGate{Analyze Gate Review}
    A2 --> AnalyzeGate
    A3 --> AnalyzeGate
    A4 --> AnalyzeGate

    AnalyzeGate -->|Pass| Improve[Improve Phase]
    AnalyzeGate -->|Fail| AnalyzeRework[Rework Analyze]
    AnalyzeRework --> Analyze

    Improve --> I1[Solution Design]
    Improve --> I2[Pilot Implementation]
    Improve --> I3[Results Analysis]
    Improve --> I4[Cost-Benefit Analysis]

    I1 --> ImproveGate{Improve Gate Review}
    I2 --> ImproveGate
    I3 --> ImproveGate
    I4 --> ImproveGate

    ImproveGate -->|Pass| Control[Control Phase]
    ImproveGate -->|Fail| ImproveRework[Rework Improve]
    ImproveRework --> Improve

    Control --> C1[Control Plan]
    Control --> C2[Standard Operating Procedures]
    Control --> C3[Monitoring System]

```

```
Control --> C4[Training Materials]

C1 --> ControlGate{Control Gate Review}
C2 --> ControlGate
C3 --> ControlGate
C4 --> ControlGate

ControlGate -->|Pass| ProjectClosure([Project Closure])
ControlGate -->|Fail| ControlRework[Rework Control]
ControlRework --> Control

ProjectClosure --> LessonsLearned[Document Lessons Learned]
LessonsLearned --> KnowledgeTransfer[Knowledge Transfer]
KnowledgeTransfer --> ContinuousMonitoring[Continuous Monitoring]
```

## Document Processing Pipeline

```
sequenceDiagram
    participant User
    participant API as TypeScript API
    participant Parser as Python Parser
    participant Analyzer as Analysis Engine
    participant Viz as Visualization System
    participant DMAIC as DMAIC Pipeline
    participant GitHub as GitHub Integration
    participant Output as Output Generator

    User->>API: Upload Document
    API->>Parser: Process Document

    Parser->>Parser: Extract Content
    Parser->>Parser: Parse Sections
    Parser->>Parser: Identify RTM Elements
    Parser->>Parser: Extract OTC Elements
    Parser->>Parser: Catalog DEL Items

    Parser->>Analyzer: Send Parsed Data
    Analyzer->>Analyzer: Enhance Elements
    Analyzer->>Analyzer: Build Relationships
    Analyzer->>Analyzer: Calculate Confidence

    Analyzer->>Viz: Generate Visualizations
    Viz->>Viz: Create Dashboards
    Viz->>Viz: Generate Charts
    Viz->>Viz: Export Graphics

    Analyzer->>DMAIC: Initialize Pipeline
    DMAIC->>DMAIC: Execute Phase
    DMAIC->>DMAIC: Track Compliance
    DMAIC->>DMAIC: Generate Reports

    Viz->>Output: Visualization Data
    DMAIC->>Output: Compliance Data
    Analyzer->>Output: Analysis Results

    Output->>Output: Generate Section 9
    Output->>Output: Create Markdown
    Output->>Output: Generate PDF
    Output->>Output: Create HTML

    Output->>GitHub: Commit Results
    GitHub->>GitHub: Create Feature Branch
    GitHub->>GitHub: Generate Pull Request
    GitHub->>GitHub: Run CI/CD Pipeline

    GitHub->>API: Webhook Notification
    API->>User: Processing Complete
```

## Compliance Tracking Workflow

```
stateDiagram-v2
    [*] --> InitialAssessment

    InitialAssessment --> DefineCompliance: Setup Metrics
    DefineCompliance --> MeasureBaseline: Establish Baseline
    MeasureBaseline --> AnalyzeGaps: Identify Gaps
    AnalyzeGaps --> ImproveProcesses: Implement Solutions
    ImproveProcesses --> ControlMonitoring: Monitor & Control

    DefineCompliance --> NonCompliant: Metrics Fail
    MeasureBaseline --> NonCompliant: Baseline Issues
    AnalyzeGaps --> NonCompliant: Critical Gaps
    ImproveProcesses --> NonCompliant: Implementation Fails

    NonCompliant --> CorrectiveAction: Initiate Corrective Action
    CorrectiveAction --> DefineCompliance: Redefine Approach
    CorrectiveAction --> MeasureBaseline: Remeasure
    CorrectiveAction --> AnalyzeGaps: Reanalyze
    CorrectiveAction --> ImproveProcesses: Reimprove

    ControlMonitoring --> Compliant: All Metrics Pass
    Compliant --> ContinuousImprovement: Maintain Standards
    ContinuousImprovement --> ControlMonitoring: Regular Review

    ControlMonitoring --> PartiallyCompliant: Some Metrics Fail
    PartiallyCompliant --> MinorCorrection: Address Issues
    MinorCorrection --> ControlMonitoring: Resume Monitoring

    Compliant --> [*]: Project Complete
```

## GitHub Integration Workflow

---

```
gitgraph
  commit id: "Initial Setup"
  branch feature/enhanced-automation
  checkout feature/enhanced-automation
  commit id: "Add Parser Engine"
  commit id: "Add Visualization System"
  commit id: "Add DMAIC Pipeline"
  commit id: "Add TypeScript Orchestration"
  commit id: "Add Configuration Management"
  commit id: "Add Section 9 Generator"
  commit id: "Add Workflow Diagrams"

  checkout main
  merge feature/enhanced-automation
  commit id: "Release v2.0.0"

  branch hotfix/compliance-update
  checkout hotfix/compliance-update
  commit id: "Fix Compliance Metrics"

  checkout main
  merge hotfix/compliance-update
  commit id: "Hotfix v2.0.1"

  branch feature/advanced-analytics
  checkout feature/advanced-analytics
  commit id: "Add ML Analytics"
  commit id: "Add Predictive Models"

  checkout main
  merge feature/advanced-analytics
  commit id: "Release v2.1.0"
```

## System Component Interaction

```

C4Component
    title Component Diagram - Enhanced Document Management System

    Container_Boundary(api, "API Layer") {
        Component(orchestration, "TypeScript Orchestration Server", "Node.js/Express",
"Handles API requests, webhooks, and real-time communication")
        Component(auth, "Authentication Service", "JWT", "Manages user authentication
and authorization")
        Component(websocket, "WebSocket Handler", "Socket.IO", "Provides real-time up-
dates and notifications")
    }

    Container_Boundary(processing, "Processing Layer") {
        Component(parser, "Document Parser Engine", "Python", "Extracts and parses RTM,
OTC, DEL elements")
        Component(analyzer, "Analysis Engine", "Python", "Enhances elements with rela-
tionships and confidence scores")
        Component(dmaic, "DMAIC Pipeline", "Python", "Manages DMAIC phases and compli-
ance tracking")
    }

    Container_Boundary(visualization, "Visualization Layer") {
        Component(viz_engine, "Visualization Engine", "Plotly/Python", "Generates in-
teractive dashboards and charts")
        Component(section9, "Section 9 Generator", "Python/Jinja2", "Creates publica-
tion-ready reports")
        Component(dashboard, "Dashboard Service", "React/TypeScript", "Serves interact-
ive web dashboards")
    }

    Container_Boundary(integration, "Integration Layer") {
        Component(github, "GitHub Integration", "GitHub API", "Manages repository oper-
ations and PR creation")
        Component(webhook, "Webhook Processor", "Node.js", "Processes external system
webhooks")
        Component(notification, "Notification Service", "Multi-channel", "Sends alerts
and updates")
    }

    Container_Boundary(storage, "Storage Layer") {
        Component(file_storage, "File Storage", "Local/Cloud", "Stores documents and
generated artifacts")
        Component(config_mgmt, "Configuration Management", "YAML/JSON", "Manages system
and workflow configurations")
        Component(cache, "Cache Layer", "Redis/Memory", "Caches frequently accessed
data")
    }

    Rel(orchestration, parser, "Initiates processing")
    Rel(parser, analyzer, "Sends parsed data")
    Rel(analyzer, viz_engine, "Provides analysis results")
    Rel(analyzer, dmaic, "Triggers DMAIC phases")
    Rel(viz_engine, section9, "Supplies visualization data")
    Rel(orchestration, github, "Manages repository operations")
    Rel(webhook, orchestration, "Forwards webhook events")
    Rel(orchestration, websocket, "Broadcasts updates")
    Rel(all, file_storage, "Reads/writes files")
    Rel(all, config_mgmt, "Loads configuration")

```

## Data Flow Architecture

---

```

flowchart LR
    subgraph Input ["Input Sources"]
        DOC[DOCX Documents]
        CONFIG[Configuration Files]
        WEBHOOK[External Webhooks]
    end

    subgraph Processing ["Processing Pipeline"]
        PARSE[Document Parser]
        ANALYZE[Analysis Engine]
        ENHANCE[Enhancement Layer]
    end

    subgraph Intelligence ["Intelligence Layer"]
        ML[Machine Learning]
        NLP[Natural Language Processing]
        PATTERN[Pattern Recognition]
    end

    subgraph Output ["Output Generation"]
        VIZ[Visualizations]
        REPORTS[Reports]
        DASHBOARDS[Dashboards]
    end

    subgraph Integration ["External Integration"]
        GITHUB[GitHub]
        JIRA[Jira]
        CONFLUENCE[Confluence]
        SLACK[Slack]
    end

    subgraph Storage ["Data Storage"]
        FILES[File Storage]
        CACHE[Cache Layer]
        LOGS[Audit Logs]
    end

    DOC --> PARSE
    CONFIG --> PARSE
    WEBHOOK --> PARSE

    PARSE --> ANALYZE
    ANALYZE --> ENHANCE

    ENHANCE --> ML
    ENHANCE --> NLP
    ENHANCE --> PATTERN

    ML --> VIZ
    NLP --> REPORTS
    PATTERN --> DASHBOARDS

    VIZ --> GITHUB
    REPORTS --> JIRA
    DASHBOARDS --> CONFLUENCE
    REPORTS --> SLACK

    PARSE --> FILES
    ANALYZE --> CACHE
    ENHANCE --> LOGS

```

## Deployment Architecture

```
deployment
  node "Development Environment" {
    artifact "Source Code"
    artifact "Unit Tests"
    artifact "Local Configuration"
  }

  node "CI/CD Pipeline" {
    artifact "Build Process"
    artifact "Integration Tests"
    artifact "Quality Gates"
    artifact "Security Scans"
  }

  node "Staging Environment" {
    artifact "Staging Deployment"
    artifact "End-to-End Tests"
    artifact "Performance Tests"
    artifact "User Acceptance Tests"
  }

  node "Production Environment" {
    component "Load Balancer"
    component "API Gateway"

    node "Application Cluster" {
      component "Orchestration Service"
      component "Processing Service"
      component "Visualization Service"
    }

    node "Data Layer" {
      database "Document Storage"
      database "Configuration DB"
      database "Cache Layer"
    }

    node "Monitoring" {
      component "Logging Service"
      component "Metrics Collection"
      component "Alert Manager"
    }
  }

  node "External Services" {
    component "GitHub API"
    component "Notification Services"
    component "Backup Storage"
  }
```